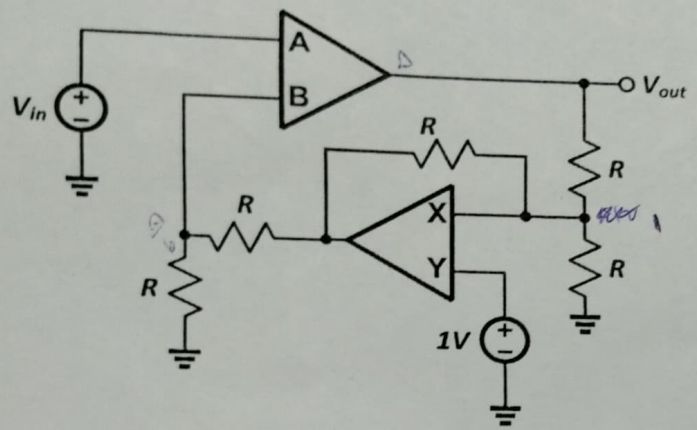
Quiz-I (Jan-May 2024)

EE2019 – Analog Systems and Lab

Total Marks: 10

Duration: 50 min

Question-1 (4 marks)



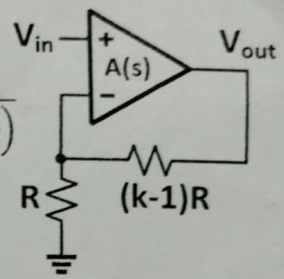
The circuit shown in the figure above operates in negative feedback configuration. Assume both opamps are ideal:

a) Determine the signs of the op-amp inputs (A,B and X,Y) for proper negative feedback operation. State your reasoning carefully. *(2 marks)*

b) Determine the ac gain V_{out}/V_{in}. *(2 marks)*

Question-2 (6 marks)

$$A(s) = \frac{10^4}{(1+s)\left(1+\frac{s}{10^3}\right)}$$



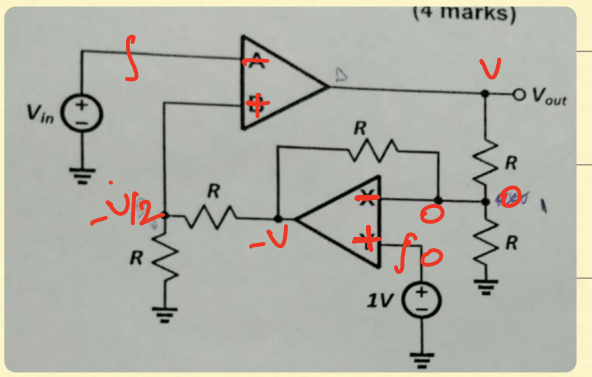
In the above circuit, assuming a non-ideal opamp of transfer function A(s), answer following questions:

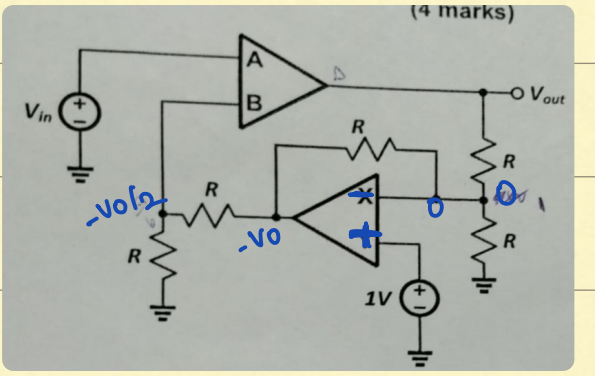
a) Comment on the stability of the system for k=1 with the help of loop gain magnitude and phase plot. *(2 marks)*

b) Determine the value of k for 60° phase margin and compare the loop gain response with k=1 plotted in (a). *(2 marks)*

c) Show the approximate location of closed loop poles on s-plane for (a) and (b). *(2 marks)*

-------X-X-X-X-X-X-X-X-X-X--------



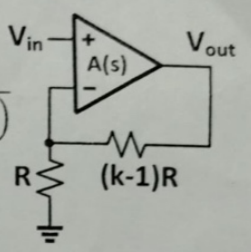


$V_{in} = -\frac{V_o}{2}$

$\Rightarrow \frac{V_o}{V_{in}} = -2$

Question-2 (6 marks)

$$A(s) = \frac{10^4}{(1+s)\left(1+\frac{s}{10^3}\right)}$$



In the above circuit, assuming a non-ideal opamp of transfer function A(s), answer following questions:

a) Comment on the stability of the system for k=1 with the help of loop gain magnitude and phase plot. *(2 marks)*

b) Determine the value of k for 60° phase margin and compare the loop gain response with k=1 plotted in (a). *(2 marks)*

c) Show the approximate location of closed loop poles on s-plane for (a) and (b). *(2 marks)*

$$LG = \frac{10^4}{K(1+s)(1+s/10^3)}$$

a) $K=1 \Rightarrow |LG| = \frac{10^4}{\sqrt{(1+\omega^2)\left(1+\frac{\omega^2}{10^6}\right)}}$

$1 \ll \omega \ll 10^3 \rightarrow |LG| = \frac{10^4}{\omega}$

$\omega_u = 10^4 > 10^3$ after 2nd pole

$-\tan^{-1}(10^4/1) - \tan^{-1}(10^4/10^3) = -174^\circ$

$PM = 180 - 174 = 6^\circ \rightarrow$ unstable

b) $PM = 60^\circ \Rightarrow -\tan^{-1}(\omega) - \tan^{-1}(\omega/1000) = -120^\circ$

$\Rightarrow \tan^{-1}(\omega) + \tan^{-1}(\omega/1000) = 120^\circ$

$\omega_u = 577.35$ rad/s $\quad \frac{10^4}{K\omega} = 1 \Rightarrow K = \frac{10^4}{\omega} = 17.32$

c)